RTL INNOVATE HACKFEST

Design a smart 3 floor elevator controller.

TEAM NAME: **VTX (Verilog vortex)**

<u>Members:</u>

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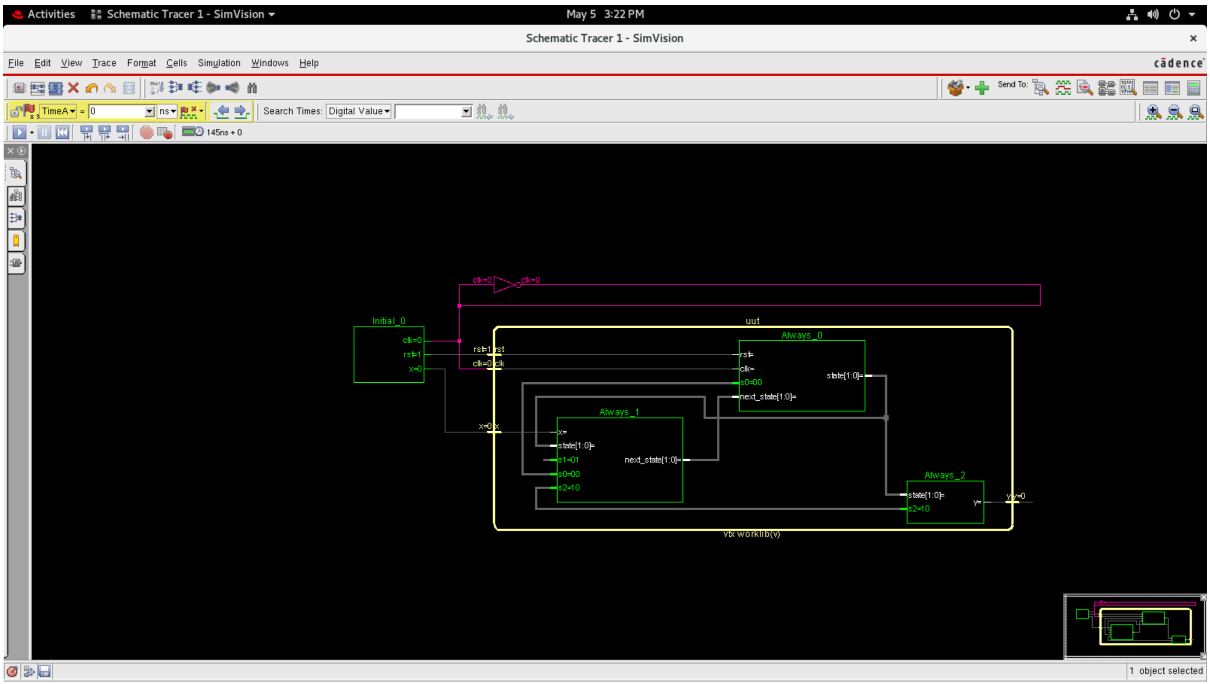
Problem statement :

Design an elevator controller for 3 floor which handles the request, controls the direction and the door timings using FSM(finite State Machine).

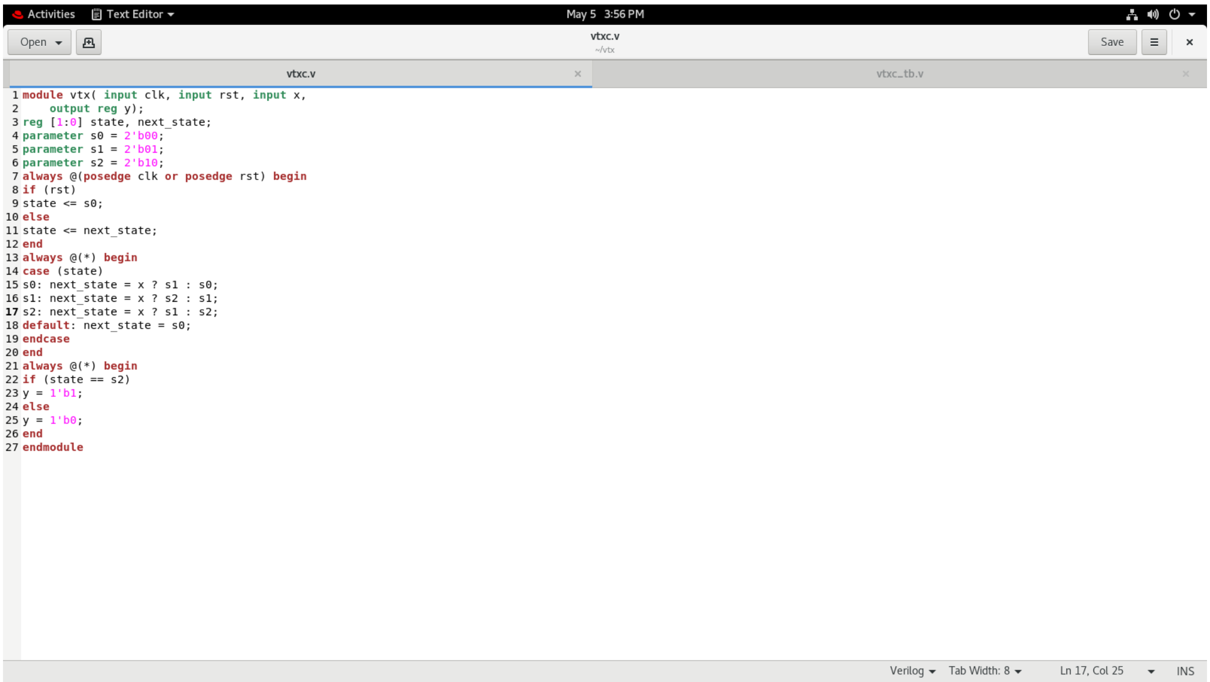
Approach:

Using Moore FSM, we have tried to implement a smart elevator controller for 3 floor,

Block diagram:



Code explanation:



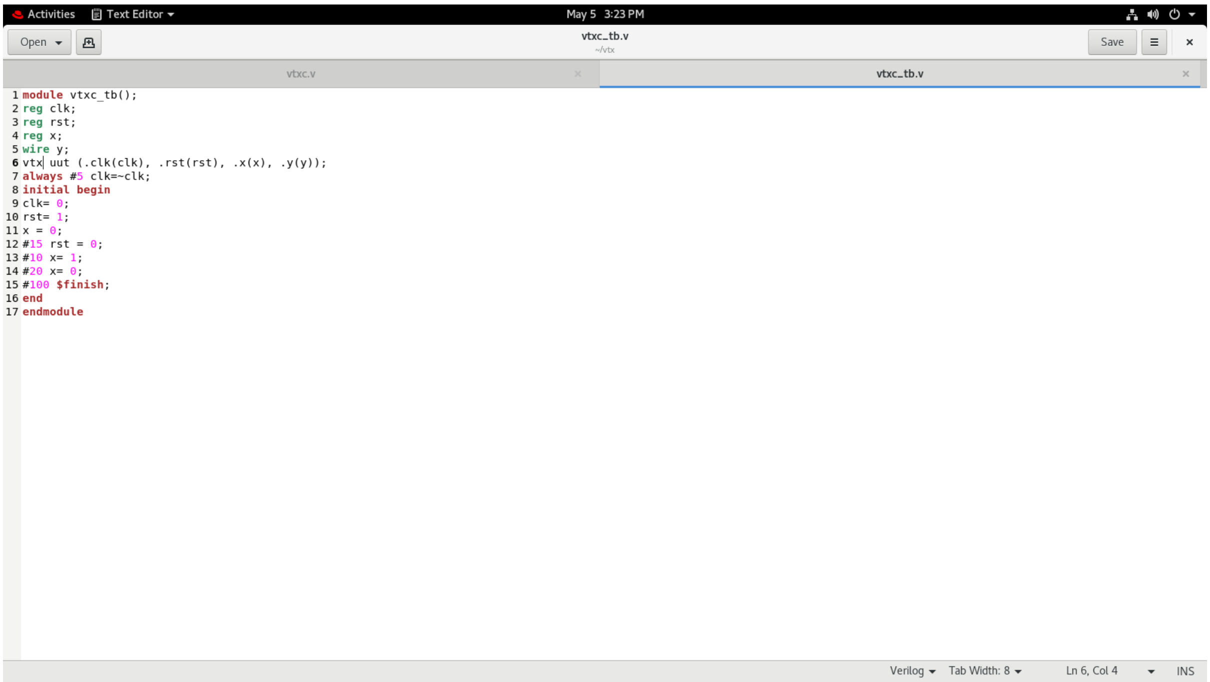

```
module vtx( input clk, input rst, input x,
    output reg y);
reg [1:0] state, next_state;
parameter s0 = 2'b00;
parameter s1 = 2'b01;
parameter s2 = 2'b10;
always @(posedge clk or posedge rst) begin
if (rst)
state <= s0;
else
state <= next_state;
end
always @(*) begin
case (state)
s0: next_state = x ? s1 : s0;
s1: next_state = x ? s2 : s1;
s2: next_state = x ? s1 : s2;
default: next_state = s0;
endcase
end
always @(*) begin
if (state == s2)
y = 1'b1;
else
y = 1'b0;
end
endmodule
```

We took 3 parameters for 3 floors(floor 0, floor1, and floor 2).

When reset(rst) is high, it denotes IDLE state in our case, which means we have no pending requests.

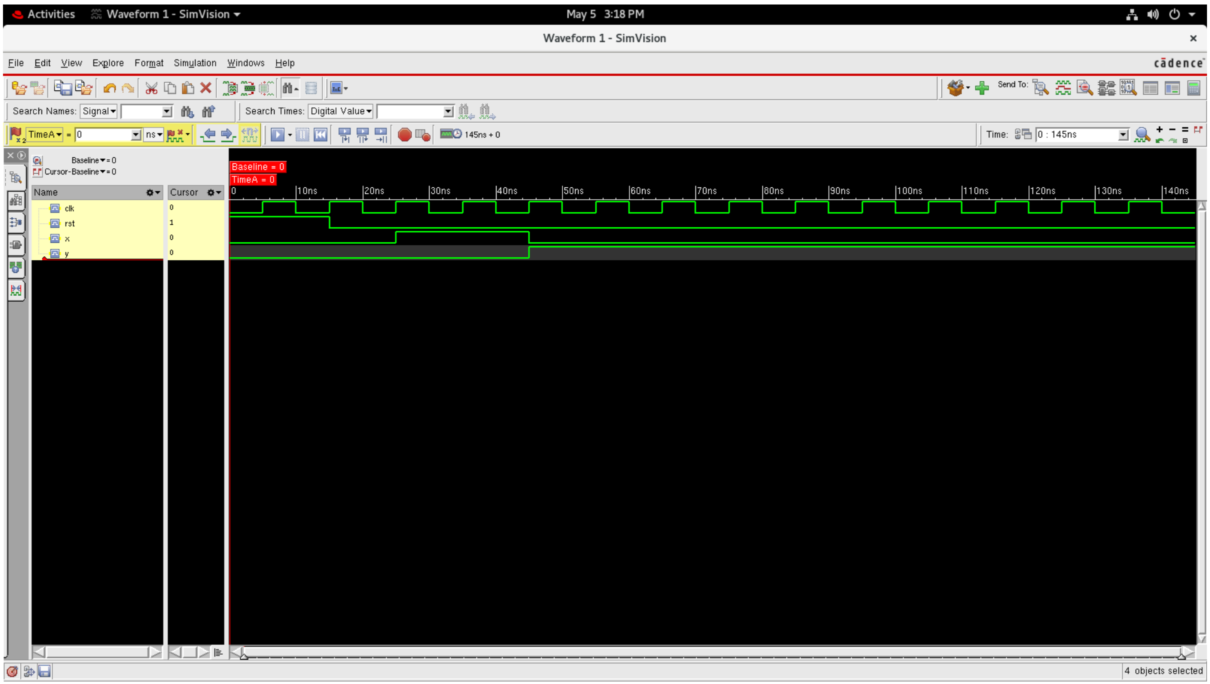
Next, for each state, when the input is high, it works as a up counter i.e. the elevator moves to the next floor else remains on the same floor. BUT in the case of 2nd floor, when the input is high, it acts as a down counter i.e. the elevator moves to the first floor otherwise remains on the same.

Testbench details:



```
module vtxc_tb();
reg clk;
reg rst;
reg x;
wire y;
vtx uut (.clk(clk), .rst(rst), .x(x), .y(y));
always #5 clk=~clk;
initial begin
clk= 0;
rst= 1;
x = 0;
#15 rst = 0;
#10 x= 1;
#20 x= 0;
#100 $finish;
end
endmodule
```

Waveform analysis:



Challenges faced:

Code debugging.

Understanding the use of blocking and non-blocking statements.